DATASCI 406 FA 26, Quiz 1.3

Name and Uniqname:

Question 1 (1 point)

To estimate the bias of $\hat{\theta}$ you simulate k datasets of size n and run `t.test(theta_hats - theta)`. You rerun with a much larger k, leaving n unchanged. What gets smaller? **Circle one.**

1. $\mathrm{Var}\left(\hat{\theta}\right)$ — the estimator itself becomes more precise.
2. The `t.test` interval around the estimated bias — we pin the bias down better.
3. $\mathrm{Bias}(\hat{\theta})$ — the estimator itself becomes less biased.
4. All three: a longer run improves every number the simulation reports.

Solution

Answer: 2. $\mathrm{Var}\left(\hat{\theta}\right)$ and $\mathrm{Bias}(\hat{\theta})$ are properties of the estimator at sample size n, and k does not touch them, which rules out 1, 3 and 4. The estimated bias is a sample mean over the k replications, with variance $\mathrm{Var}\left(\hat{\theta}\right)/k$, so a larger k narrows the interval around it. Raising k tells you more about the estimator; only raising n changes the estimator.

Question 2 (1 point)

You simulate to check whether `t.test(x)$conf.int` really covers 95% of the time for skewed data. Which ingredient does the simulation need that a real data analysis can never supply? **Circle one.**

1. The true value of θ, so we can check whether the interval caught it.
2. A sample size n matching the dataset we actually have.
3. Enough replications k that the reported coverage is precise.
4. A confidence level $1-\alpha$ fixed before looking at the data.

Solution

Answer: 1. Coverage is $\mathrm{E}\left(\mathbf{1}(A \le \theta \le B)\right)$, and you cannot evaluate that indicator without knowing θ. Setting the truth is what makes this a simulation. The other three are choices you make in a real data analysis too.

Question 3 (3 points)

$X_1, \ldots, X_{10}$ are i.i.d. $N(\mu, \sigma^2)$ with $\sigma^2 = 1$. Two estimators of σ^2, run on the same $k = 100{,}000$ simulated datasets:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \qquad \hat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

	bias	variance
S^2	0.00	0.22
MLE	-0.10	0.18

Circle the estimator you would use and justify it in two or three sentences. You must say **both** (i) which single quantity ranks the two, and why neither column settles it alone, and (ii) what that quantity is for each estimator.

Circle: $\mathbf{S^2}$ or **MLE**

Solution

Answer: MLE.

1. **The criterion.** Mean squared error, which charges an estimator for both columns: MSE = variance + bias^2. Bias alone would pick S^2 and variance alone would pick the MLE, so neither column settles it.
2. **The arithmetic.** S^2: $0.22 + 0.00 = 0.22$. MLE: $0.18 + 0.01 = 0.19$. The biased estimator wins.

The exact values are $2/(n-1) = 0.222$ and $2(n-1)/n^2 + 1/n^2 = 0.190$. Both numbers in the table are themselves Monte Carlo estimates, and a comparison like this can depend on n, though here the MLE wins at every n.

Rubric. 1 pt for MLE, 1 pt for naming MSE and variance + bias^2, 1 pt for the two numbers. Saying only that a biased estimator can sometimes win does not earn the last two points; the comparison has to be made.